

NATIONAL UNIVERSITY OF SINGAPORE

Department of Statistics and Data Science

ST3131 Regression Analysis

(Semester 2 : AY 2025/2026)

ASSIGNMENT

Due Date: Friday, 10 April 2026, by 12 noon

INSTRUCTIONS TO STUDENTS

1. Students are supposed to submit your work on time. Any submission after the due time of the due date are marked as late.
2. **5% of the given mark will be deducted for each 1 hours late in submission.**
3. **No extension on the deadline.**
4. Students are required to complete this assignment individually.
5. Your submission has **two separate files**. One is a .pdf file of the report, and the second file is of the code file (.R file, or .Rmd file or .py file or .ipynb file). Make sure that there is no error when the graders open and run your code file.
6. Be sure to lay out systematically the various parts and steps in your report as well as in your code file.
7. Student must declare if any LLM/AI is used to help generating the report. Student's original work must contribute at least 50% of the report if any LLM/AI is used.

The price of a HDB resale flat in Singapore depends on many factors. The data given in the file `hdb-resale-July-2020.csv` concern the selling price of HDB resale flats to some variables given in the data in July 2020. This data set is extracted from published website.¹ Description of the columns in the file is given in Table 1.

Purpose: propose a model that you think it is the best among all the possible linear models that helps to predict the price of resale flats.

You may use WLS, GWLS or Ridge regression, but Principal Component Regression should not be presented.

¹<https://data.gov.sg/dataset/resale-flat-prices>

Variable	Description
month	the month when the flat was sold (July 2020)
town	the town where the flat belongs to
flat_type	the type of flat
block	the name of the block (in number) that the flat belongs to
street_name	name of the street on the address of the flat
storey_range	range of the storey where the flat is at
floor_area_sqm	the area of the flat's floor in square meters
flat_model	model name of the flat
lease_commence_date	the year when the lease of the flat started ^(*)
remaining_lease	the remaining lease at the time the flat was sold
resale_price	resale price of flat, in SGD

Table 1: Variable description. (*) For Singapore HDB flats, the lease is limited to 99 years.

Your report can be organized in the following manner:

Part 0 Introduction

In this part, you need to introduce the problem (main goal of the report), introduce the data set and the steps that you plan to follow in the subsequent parts of the report.

Part I EDA: Exploring the variables and association

What you should do:

- Summarize response variable and the possible relationship of the response variable with the regressors by descriptive statistics, figures or plots (histogram, scatter plot, box plot, normal probability plot, etc.).
- You need to describe: variable type; possible distribution of response and comment on possible relationship between variables.
- Student is suggested to self-search to have some general understanding on the HDB flats in Singapore, such as about location. That could help if you want to re-define the location of the flats in the data (instead of using the original `town` and/or `street_name`).

Part II Building Models

What you should do:

- Fit multiple linear regression model(s). Present the coefficients table and Anova table (for at least one model).
- Describe the parameter estimation significance. Give full steps of model adequacy checking and propose treatment if it is needed.
- Do a model selection to find a better model. You may provide some graphs drawn after fitting a model to see if the model fitted adequately.
- State your final model and interpret this model (interpret the effect of at least 2 regressors, one quantitative and one categorical if the model has both).

NOTES: The initial model, M_0 , should have at least six regressors. You might use the original input variables given in the data set or modify them (if needed) before adding them to the initial model. Do explain on the modification used.

After fitting M_0 , you might continue with many intermediate models before reaching the final models. However, in this report with limited pages, you don't need to present all those intermediate models. You just need to present at least two models: initial model, (M_0) and the final model of your choice (M_N).

If the final model is very different from the initial model, you should explain the flow on how you go from the initial to the final model.

Format of the report

1. Your report is a .pdf file, limited to **no more than SIX printing pages, font size 12, margins of not less than 0.75 inches.**

The standard for a typed A4 document is 1 inch for each margin.

2. Table and/or figure in the report should be numbered clearly.
3. If you submit the report without submitting the code file, your mark will be deducted by half of the mark given to your report.
4. If you add any code into your report, it will still be counted within the six pages allowed. Hence, it's advised not to add the code into your report.

Notes: This is a statistical report, hence, please present it in a report style. It should not be presented as a list of answers/solutions for a list of questions.

END OF ASSESSMENT